

Supervisor Meeting — Talking Points

Hai Nguyen · B.Sc. Computer Science and Experimental Physics, LMU Munich

Thesis: Benchmarking Tabular Data Augmentation Techniques for Deep Clustering · 10 official weeks, 15 June – 23 August 2026

1 · SELF-INTRODUCTION

CONVERSATIONAL · main version

“So, a bit about me — I'm Hai, I'm doing a combined Bachelor's in Computer Science and Experimental Physics here at LMU, and I'm now at the thesis stage. On the computer science side, I've done a software engineering internship — I built the controller and the database for a small game project — and I was a teaching assistant for the Software Engineering course. The physics side really shaped how I work: in the lab courses I did a lot of hands-on data analysis and visualization with SciPy and Matlab, error analysis, and writing everything up as proper scientific reports — so empirical work and benchmarking feel natural to me. I also have a strong maths background, which helps on the theory side, and I recently took a Data Mining course, so clustering and the evaluation metrics — NMI, ARI, silhouette — are already familiar. The one thing I'll be upfront about is that the deep-learning tooling, PyTorch and scikit-learn, is new to me — and that's exactly why I've built a preparation phase into my plan to get up to speed before the official start.”

FORMAL

“My name is Hai Nguyen. I'm a Bachelor's student at LMU in a combined Computer Science and Experimental Physics programme, currently with a 1.7 average, and I'm now at the thesis stage. My computer science experience includes a software development internship, where I implemented the controller component and the database of an MVC-based application, and a semester as a teaching assistant for the Software Engineering course. My physics background has given me solid practical experience in empirical work: in the advanced lab courses I carried out data analysis, visualization and error analysis using SciPy and Matlab, and documented the results in scientific reports. I also have a strong mathematical foundation and recent coursework in Data Mining, which covered clustering methods and the evaluation metrics central to this thesis — NMI, ARI and silhouette. I should note that deep-learning frameworks such as PyTorch and scikit-learn are new to me; I have therefore scheduled a dedicated preparation phase before the official start to build that competence.”

30-SECOND · short backup

“I'm Hai Nguyen, a Bachelor's student at LMU in a combined Computer Science and Physics programme. I have a software engineering background — an internship and a teaching assistant role — and a lot of empirical, data-analysis experience from my physics lab courses. I recently took a Data Mining course, so clustering and the evaluation metrics for this thesis are already familiar. PyTorch is new to me, but I've planned a preparation phase to get up to speed before the official start.”

POINTS TO RAISE WITH YOUR SUPERVISOR

- Scope: is a simple MLP-encoder + k-means backbone acceptable, and how many datasets / augmentations do you expect?
- Does 10 weeks part-time look realistic to you? I have a scope-reduction fallback if needed.
- Meeting cadence — I've pencilled in check-ins after week 2 and week 7.
- Compute: I plan to use Google Colab for the GPU.

2 · THESIS PLAN

CONVERSATIONAL · main version

“So for the thesis, the idea is to benchmark different tabular data augmentation techniques and see how they affect deep clustering. The setup is contrastive learning — I train an MLP encoder to pull together augmented views of the same sample, and then run k-means on the embeddings it learns. The real question is which augmentations actually help produce cluster-friendly representations, and which ones distort the data and hurt clustering. I'd evaluate that with NMI and ARI. On timing, I have the 10 official weeks from the 15th of June to the 23rd of August. Since the topic needs PyTorch and I'm coming more from a software engineering and physics background, I'm using the weeks before the official start as a preparation phase, so I don't lose official time to that. The plan itself: the first two weeks are literature review and fixing the method and datasets; weeks three to five are the implementation, and I want a minimal pipeline running end-to-end by around week four; then experiments and sensitivity studies; and the last weeks are analysis and writing. I've split it into six milestones so we can both track progress. I'd keep the scope deliberate — a simple encoder, three or four datasets, a handful of augmentations — and I wanted to check with you whether that scope feels right.”

FORMAL

“The thesis is an empirical benchmark of tabular data augmentation techniques for deep clustering. The approach is based on contrastive learning: an MLP encoder is trained so that augmented views of the same sample are pulled together, and k-means is then applied to the resulting embeddings. The research question is which augmentation techniques produce cluster-friendly representations and which degrade clustering performance — evaluated with NMI and ARI. The official processing time is ten weeks, from 15 June to 23 August, on a part-time basis. As the topic requires PyTorch, I've scheduled a preparation phase before the official start to learn the necessary tooling. The official period is structured into five phases: literature review and conception in weeks one and two, implementation in weeks three to five, experiments and sensitivity studies in weeks six and seven, analysis and writing in weeks eight and nine, and revision and submission in week ten. Progress is tracked through six milestones. I intend to keep the scope deliberately focused — a simple encoder, three to four datasets, and a small set of augmentations — and I'd appreciate your feedback on whether that scope is appropriate.”

30-SECOND · short backup

“My thesis benchmarks different tabular data augmentation techniques for deep clustering. I train an MLP encoder with contrastive learning, then cluster the embeddings with k-means, and measure quality with NMI and ARI. I have ten official weeks, from mid-June to late August, with a preparation phase beforehand to learn PyTorch. The plan goes from literature review to implementation, experiments, and writing, tracked with six milestones. I'm keeping the scope small — a simple encoder, a few datasets, a handful of augmentations — and wanted your view on whether that's the right level.”